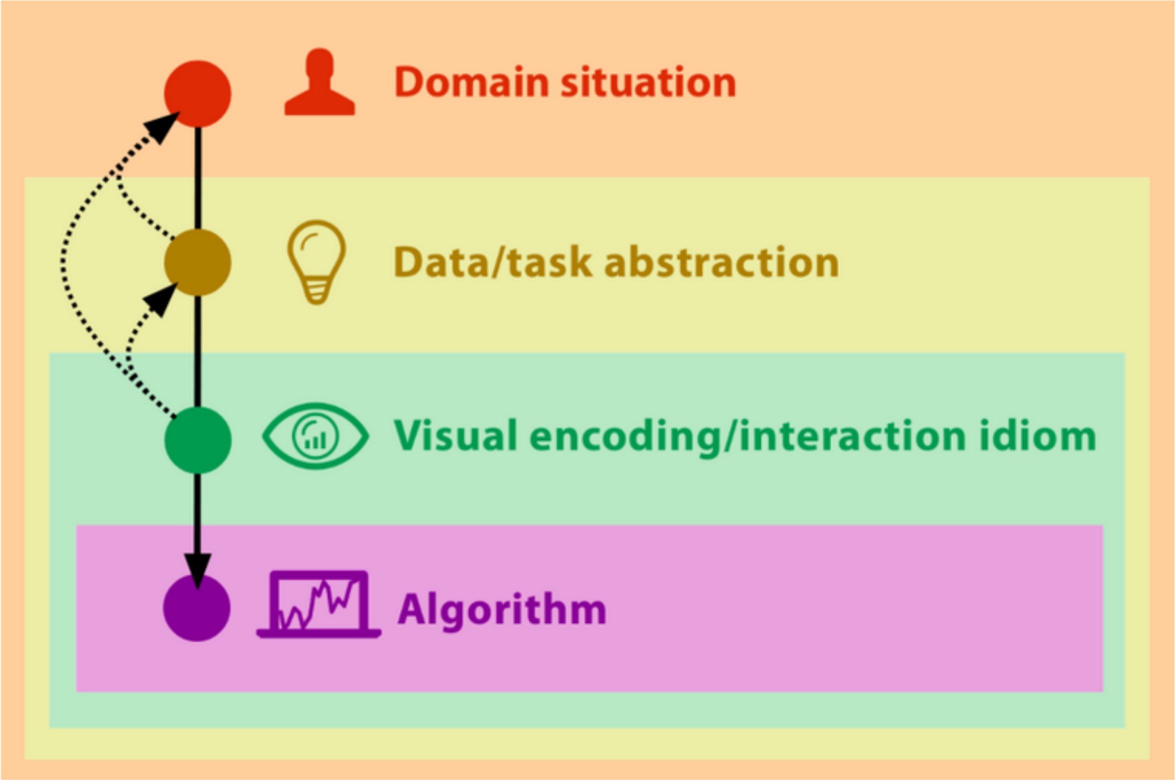
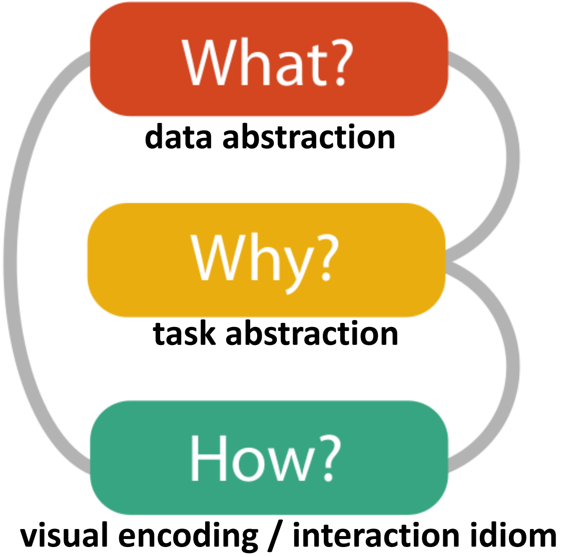
How? To visualize Color

DSK808 Data Visualization

Fall 2025

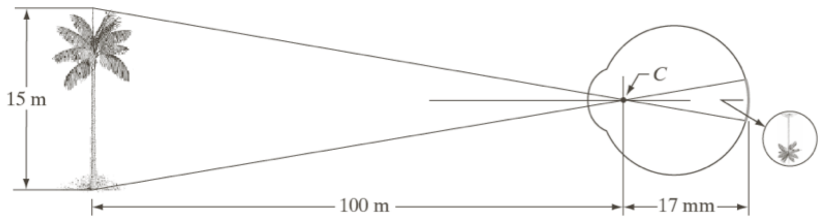
Participatory Visualization Design





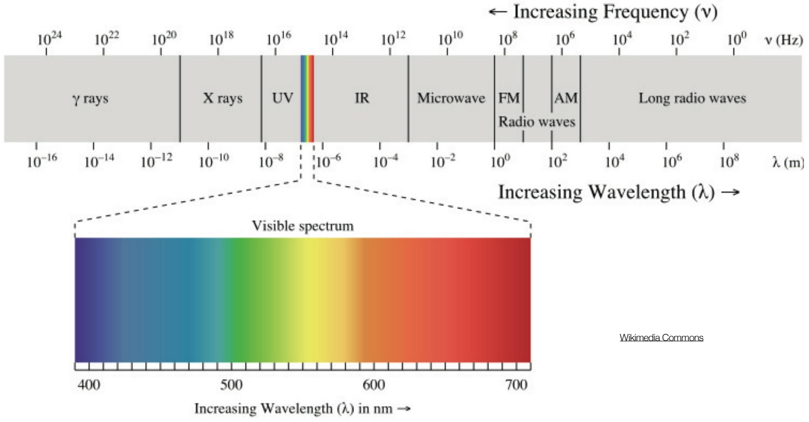
Physiology

- The main sensory component of vision involves recording of light scattered from objects of a scene, forming an inverted two-dimensional function on the photoreceptors of the eye
- Given this sampled information of incoming light a model of the environment is formed
  - Sampling over space and time



Visible Light

- Light is an electromagnetic wave with the wavelengths defining the apparent color
  - Laser light consists of a single wavelength λ
- Visible light is in the spectrum of λ = 400nm (violet-blue) to 700nm (red) perceivable by the human visual system
  - Very small section of the spectrum between x-rays, UV and microwaves, infrared (heat), radio



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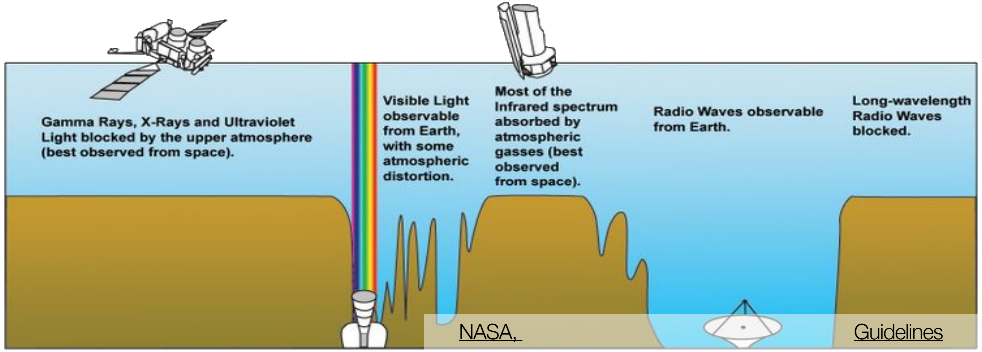
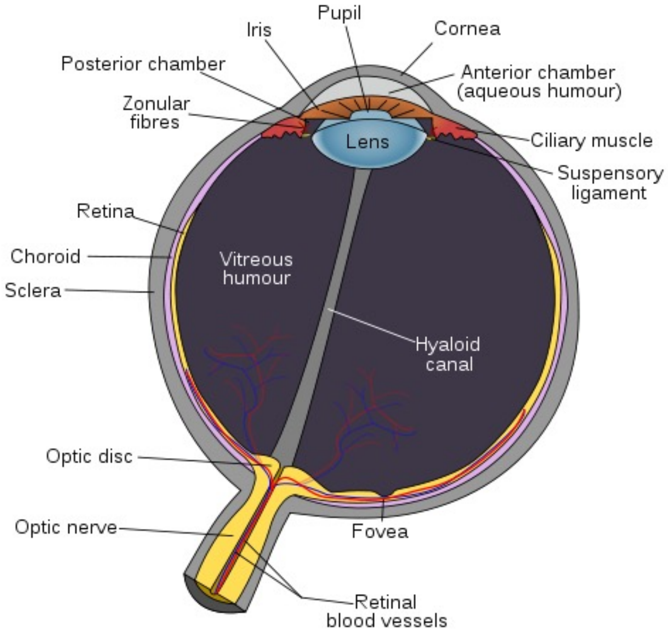


NASA, Guidelines

Anatomy of the Visual System

- Horizontal cross-section of the human eye reveals its complexity
  - Light path goes through cornea, iris, pupil, lens and retina
- The lens system is controlled to be continually oriented at a region of interest
  - Fast and continually making adjustments for image stabilization
- Optical system similar to a double-lens camera
  - Cornea focusing the light onto the main lens and protecting the internal structure
  - Pupil and iris control amount of incoming light (aperture)
  - The lens can adjust the focal length of the system (depth focus)
  - Finally light rays reach the retina



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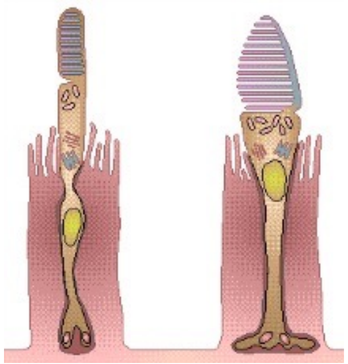
Retina

- The retina contains a layer of photoreceptors responsible for recording the incoming light ray intensities
- Two types, rods (left) and cones (right) for intensity (brightness) and color perception
  - Rods being typically ten times more sensitive to light than cones
- Distribution of photoreceptor roughly radially symmetric but not uniform over angle
  - Fovea in the center with only cones of high density for sharpest vision
    - 120 million rods and 6 million cones overall
- Significant visual preprocessing done in the eye
  - Only about 1 million optical nerves connecting to the brain



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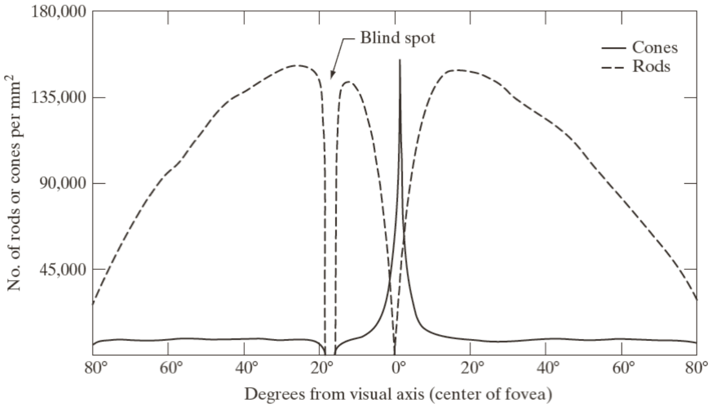
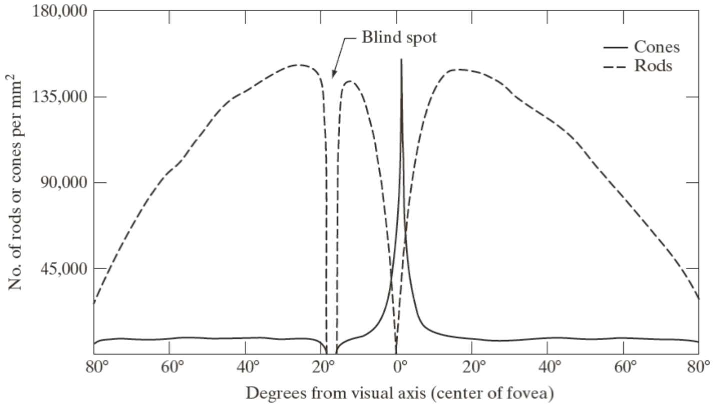
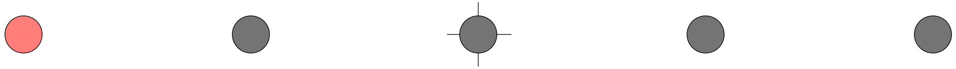


Blind Spot Experiment

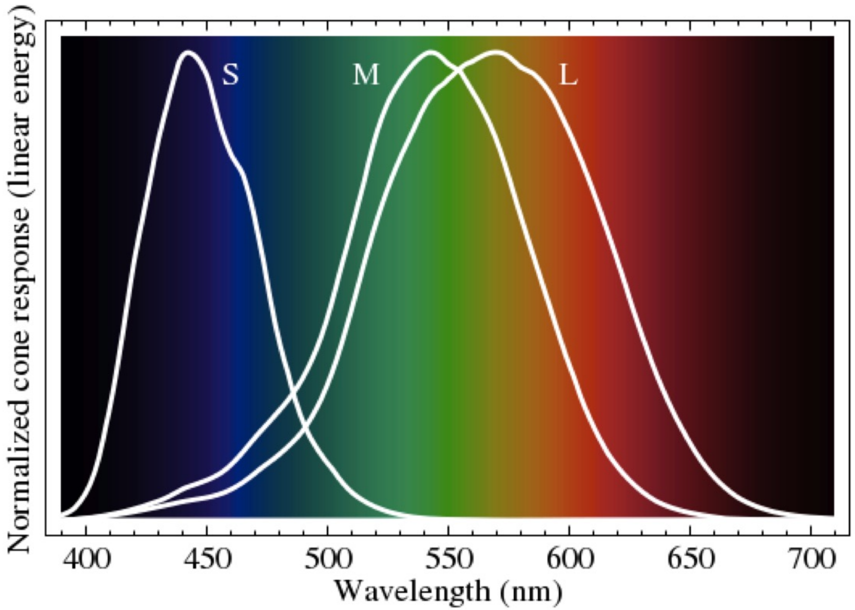
- Blind spot where optical nerves leave the eye
  - Off center at approx. 20°





Color Sensitivity

- Rods for scotopic, night vision for increased sensitivity in low light conditions
  - Only record achromatic light, sensitive to the entire spectrum
- Three types of cones for short, medium, and long wavelengths
  - Responsible for color vision
  - Much fewer *short* ones



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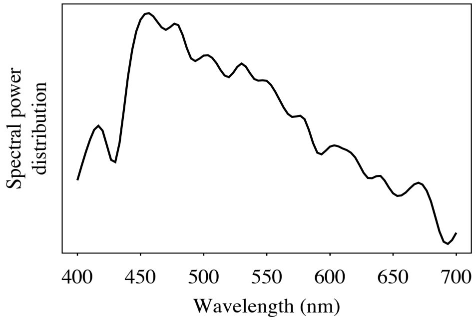
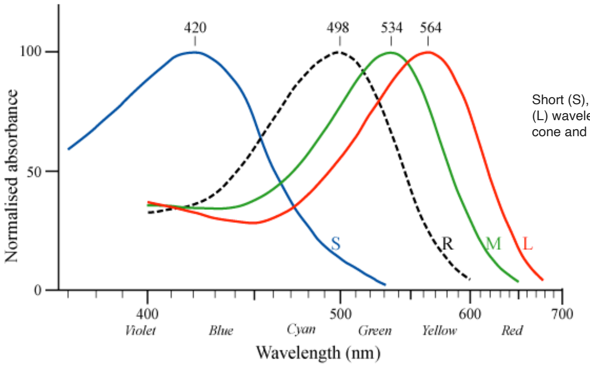
Color

- Color is the combination of different wavelength intensities
  - Integral over the spectral power distribution E(λ)
    - daylight example
  - Too complex to use as a color description
- Visual system perceives color through the different sensitivities of cones in the retina
- Trichromatic vision
  - Three types of cones respond to different spectral ranges
    - roughly red, green and blue!
    - blue sensitivity about 20 times lower



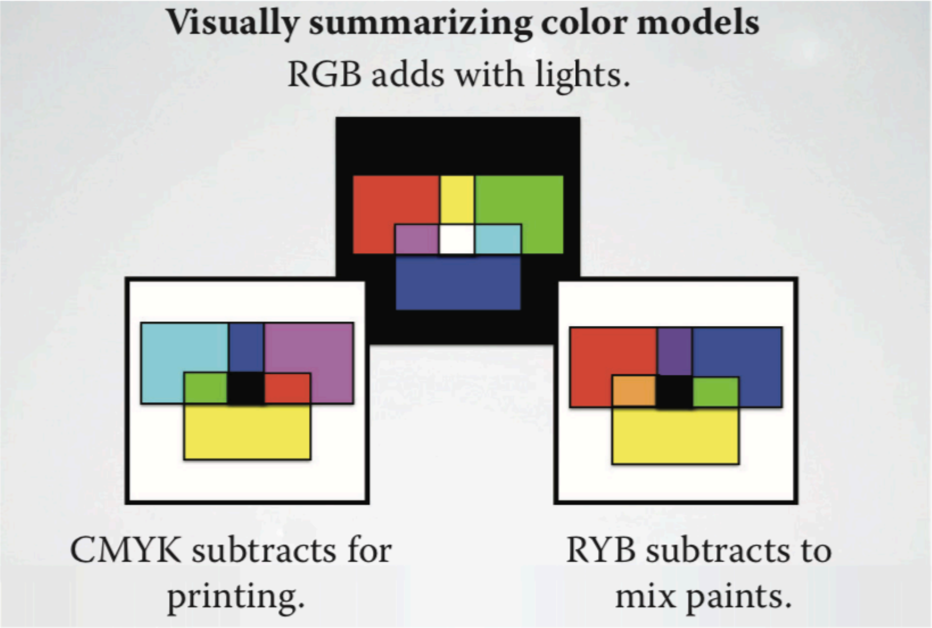


Short (S), medium (M) and long (L) wavelength pigments in human cone and rod (R) cells.

Color Models

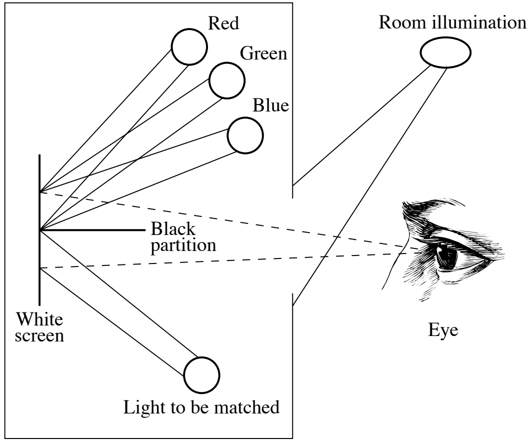
A color model is a structured system for creating a full range of colors from a small set of defined primary colors.

There are three fundamental model



Trichromacy theory

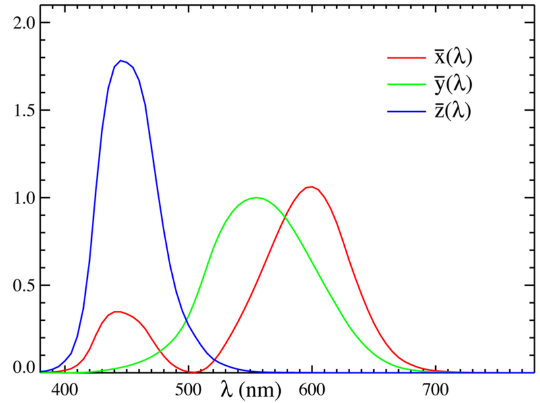
- **Thichromacy:** based on the most impact fact about our color vision: we have three distinct color receptors: cones.
- **Color Space**: three dimensional organization of color.
- **Color model**: an abstract mathematical model describing the way colors can be represented as tuples of numbers.



® Ze-Nian Li, Mark S. Drew, Fundamentals of Multimedia. Pearson Prentice Hall, 2004.

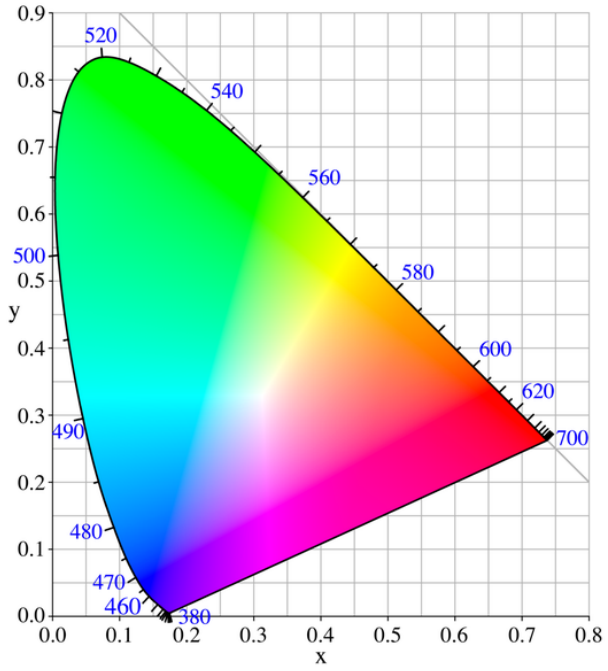
Color Models

- Tristimulus theory
  - Any color can be matched using three additive primaries
    - varying each primary color to match a given color in the color matching experiment
- RGB color matching has *negative* lobe on the $\tilde{r}()$ curve
- CIE defined fictitious color matching curves $x(), y(), z()$ resulting in chromatic *tristimulus* values $\mathrm{X}, \mathrm{Y}, \mathrm{Z}$
    - ***C**ommission **I**nternational de l'**É**clairage*
- HSV (Hue, Saturation, Value) is more closely align with the way human vision perceives color-making attributes.



CIE Chromaticity Diagram

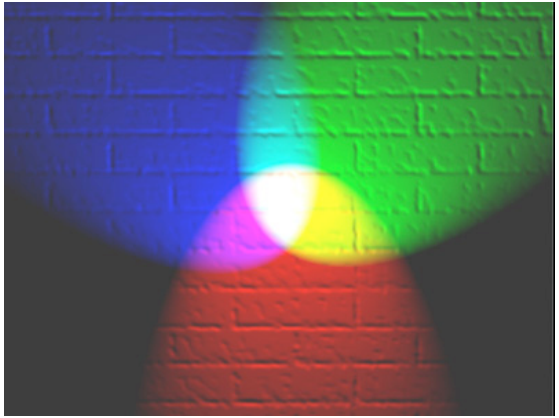
- Normalized representation in 2D color diagram
  - $x = X / (X+Y+Z)$ $y = Y / (X+Y+Z)$
  - $z = 1 - x - y$
- "Horse Shoe" outer curve is monochromatic locus
  - Pure single-wavelength light
  - Dominant wavelength is locus projection
  - Line of purples
- Equi-energy white color at (1/3,1/3)
  - Different whites defined
    - daylight, cool and warm whites …
- Mixture of lights is a linear combination
  - Light is additive



http://upload.wikimedia.org/wikipedia/commons/b/b0/CIExy1931.png

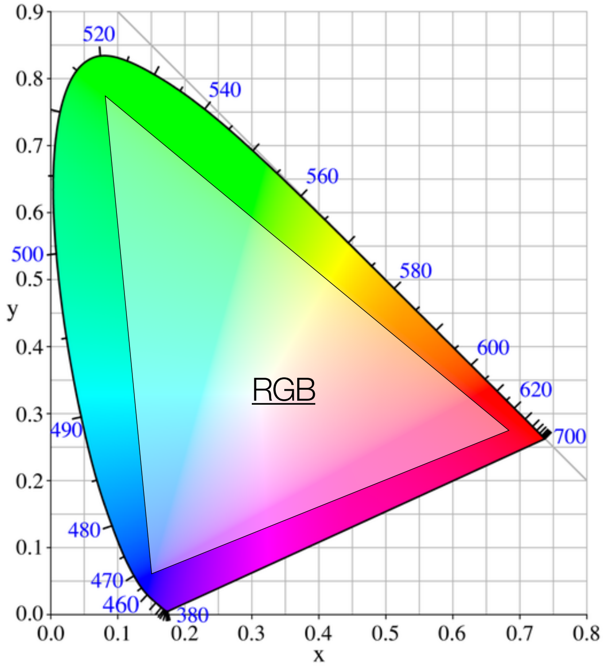
Additive RGB Color

- Linear additive combination of real RGB primary colors
  - Desired color achieved by adding together multiple light sources
  - Computer monitors, televisions, projectors …
- Similar to integral over spectral power distribution
  - Integrating over only three primary colors instead of all wavelengths
- Combinations of two RGB primaries yield secondary colors
  - Cyan, magenta, yellow



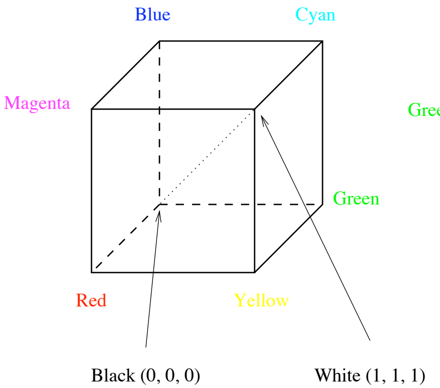
Limits of Color Models

- “Horse Shoe” area describes all possible color hues
  - CIE non-physical color model in X, Y, Z can represent all color hues, as well as brightness and saturation
- Color model with limited set of "real" primary colors has limited color space reproduction
  - **Gamut** = region of reproducible color
- RGB is good choice but is nevertheless limited
  - Cannot represent all real world colors and is not a canonical model
    - depending on output device

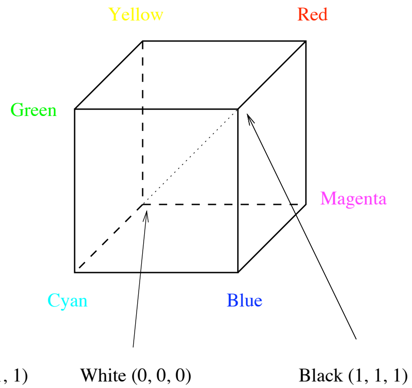


Subtractive Color

- Additive colors unsuitable for printing process
  - No addition to white (paper) possible
  - Deposit of ink must thus *remove* from white
- CMYK removes cyan, magenta and yellow from white
  - Yellow absorbs blue → reflects green and red = yellow
- Cyan, magenta, yellow primary subtractive colors (K is for black, removal of undercolor)

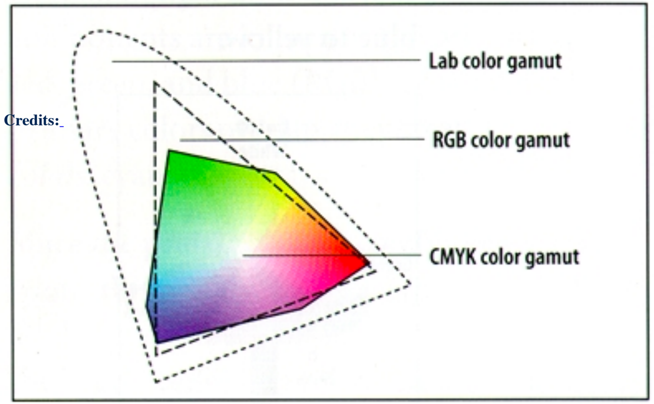


The RGB Cube



The CMY Cube

Credits:



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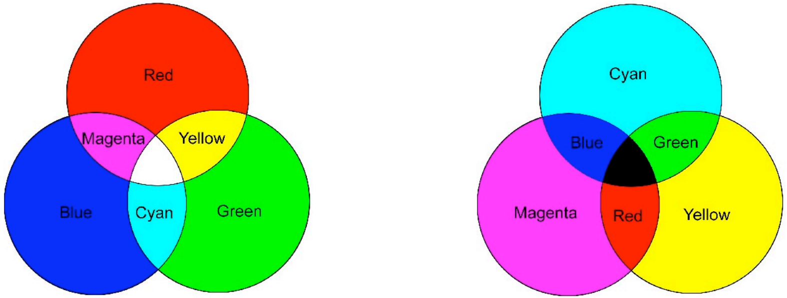
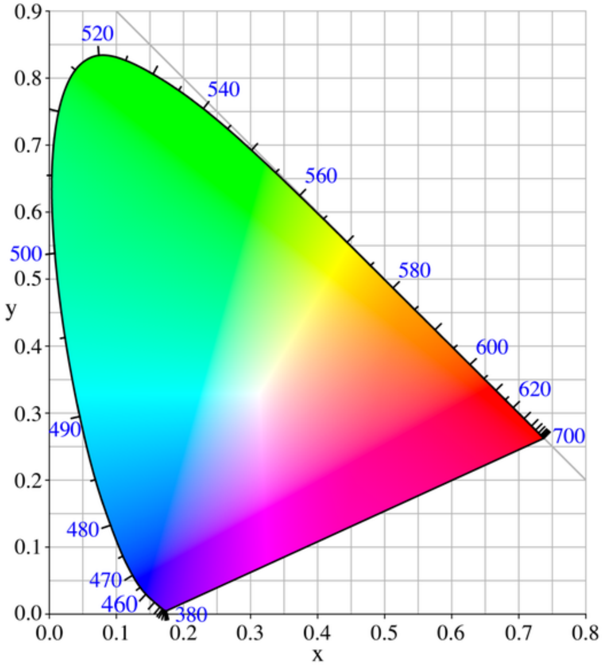


® Ze-Nian Li, Mark S. Drew, Fundamentals of Multimedia. Pearson Prentice Hall, 2004.

HSV Colors

- The intensity independent color hue can fully be defined by interpolation from two wavelengths
  - Thus requires only two *chromatic* color channels
  - The brightness (intensity) requires one additional single *achromatic* color channel



HSV Colors

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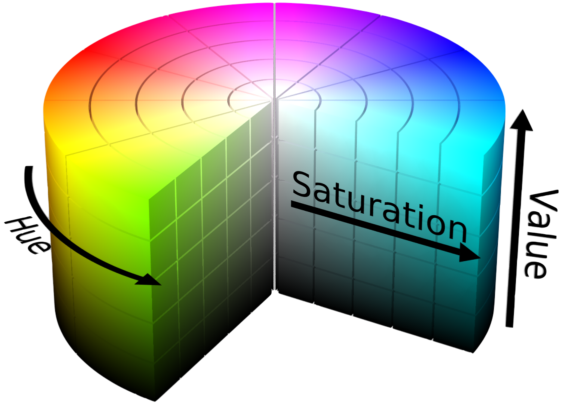
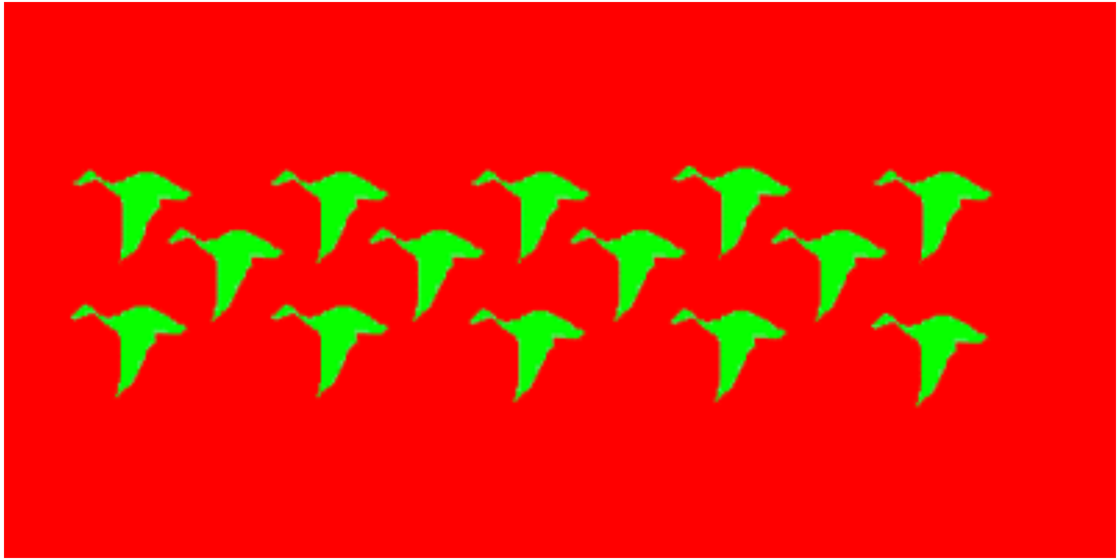
  - Thus requires only two *chromatic* color channels
  - The brightness (intensity) requires one additional single *achromatic* color channel
- ***Hue*** identifies the dominant (or its complementary) wavelength
  - Angle around vertical color space axis
- ***Saturation*** indicates strength of the color, least being white
  - Distance from center axis
- ***Value*** along the axis denotes overall brightness
- Conversion from RGB
  - Same device dependency and limits
- Perceptually not uniform
  - See later, CIELAB model variation from CIE X, Y, Z model



Aspects not covered by Trichromacy Theory

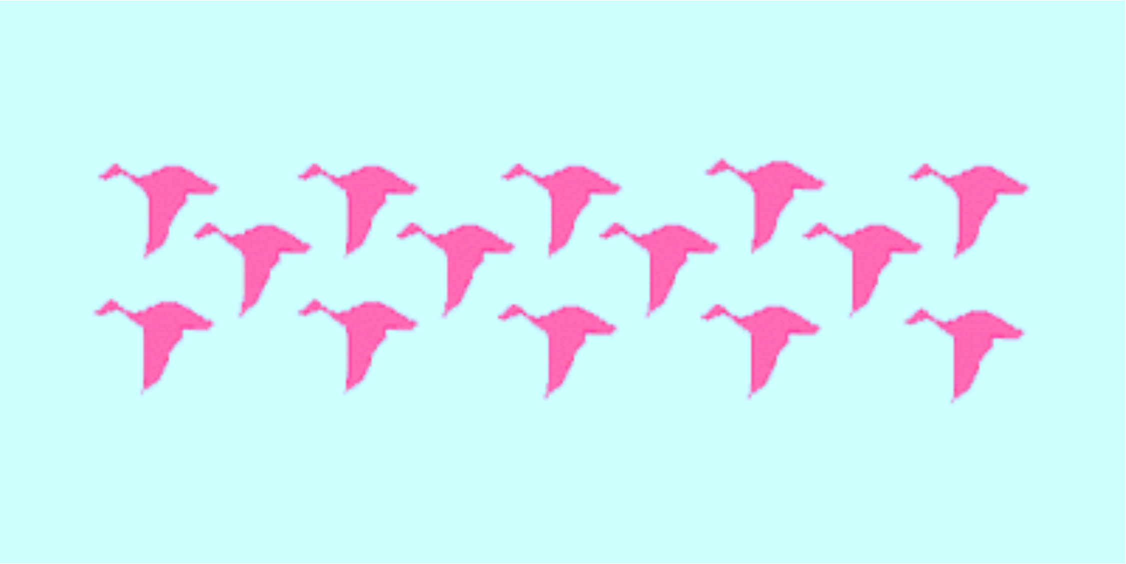
- Afterimage effect
- Missing color combinations
- Color Blindness

Afterimage Effect



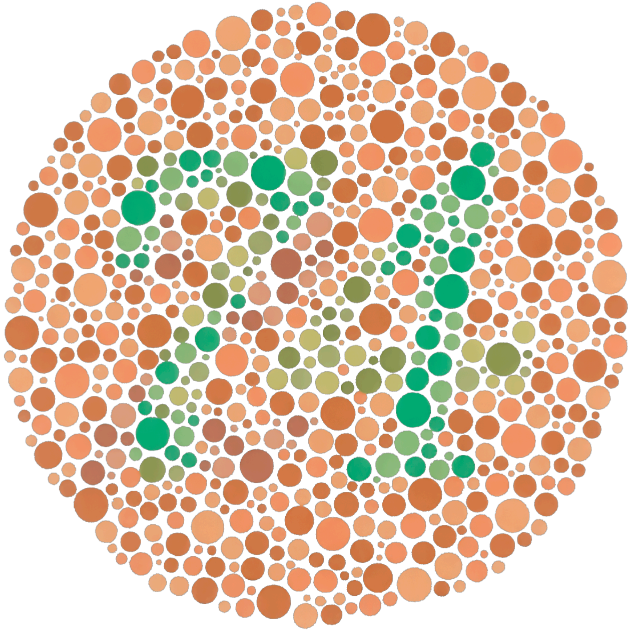
Afterimage Effect

Afterimage Effect



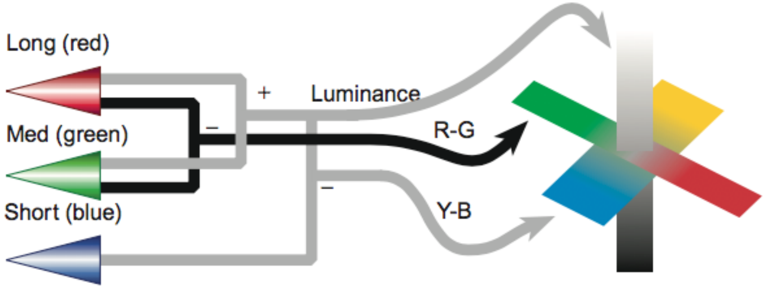
Missing color combinations

Color Blindness



Opponent Color Spaces

- Ewald Hering (1920). *Six elementary colors arranged perceptually as opponent pairs along three axes*:
  - black – white
  - red – green
  - yellow - blue



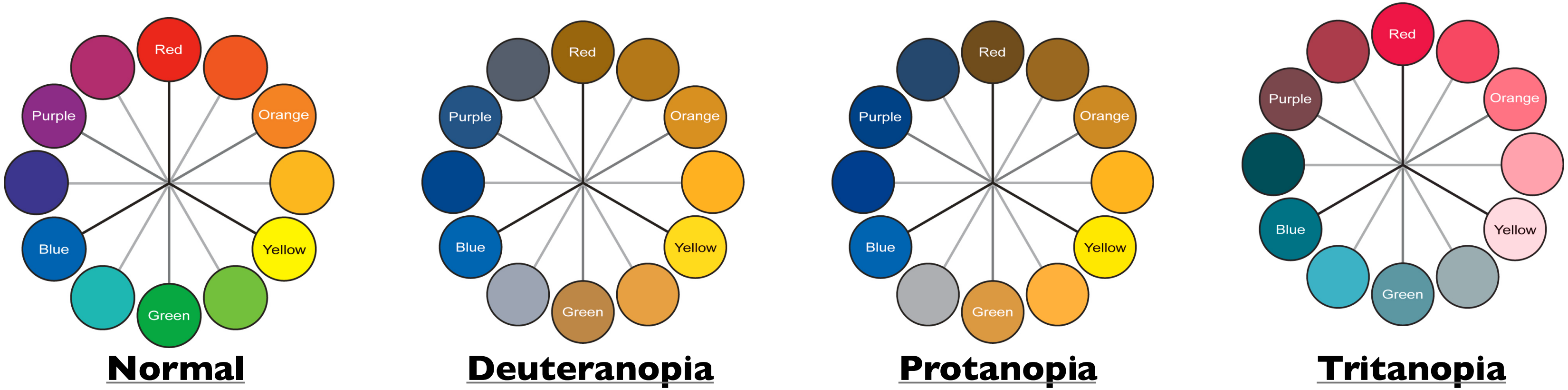
® Colin Ware. Information Visualization: Perception for Design. Morgan Kaufmann, 2013.

- Visual input is processed into three distinct channels:
  - The black–white channel (luminance) is based on input from all the cones
  - The red–green channel is based on the difference of long- and middle- wavelength cone signals
  - The yellow–blue channel is based on the difference between the short-wavelength cones and the sum of the other two

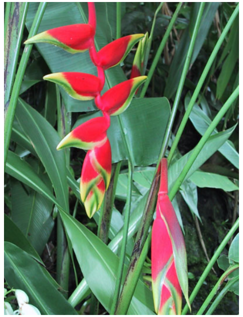
Opponent Color Spaces and Color Deficiency

- Opponent color spaces have two “chromatic” axes
- “Color blindness” means one axis has degraded acuity
  - Reduces color to 2 dimensions
  - 8% of men are red-green color deficient
  - Blue-yellow deficiency is less common

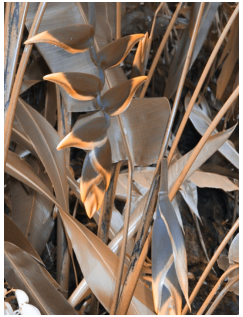
® Seriously Colorful: Advanced Color Principles & Practices. Stone.Tableau Customer Conference 2014.



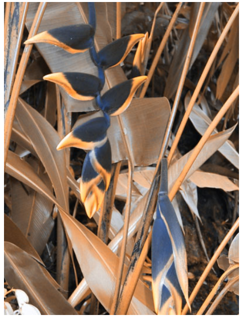
Opponent Color Spaces and Color Deficiency



Normal

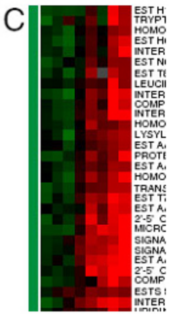


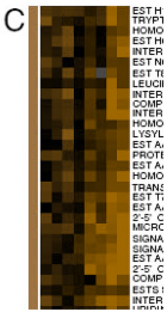
Deuteranopia



Protanopia

Tritanopia





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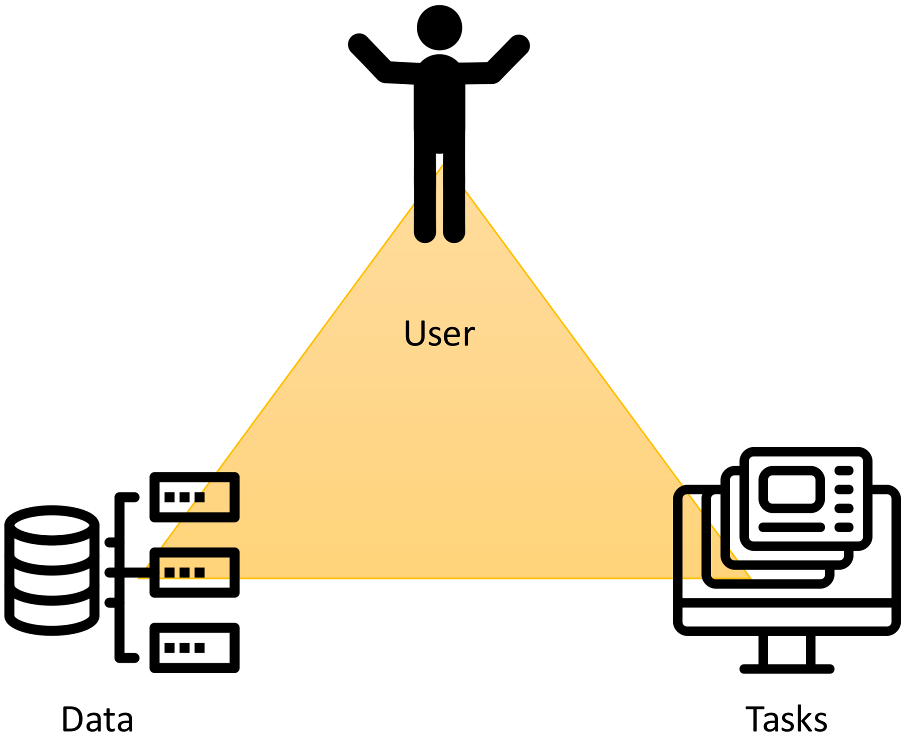
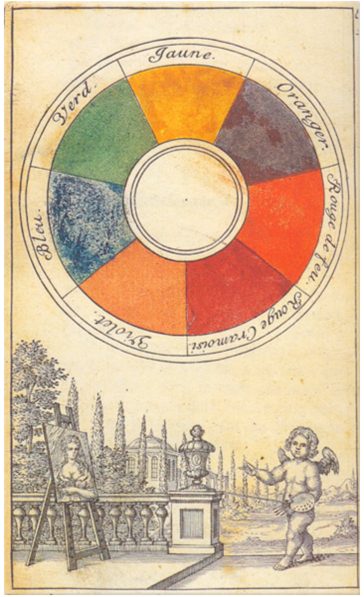
Color Design



Complementary Colors

- Newton’s arrangement of color on a circle
  - Based on the **painters** primary colors red, yellow, blue
- Arrangement of complementary colors opposite of each other
  - Complementary colors enhance the other’s effect through optical contrast

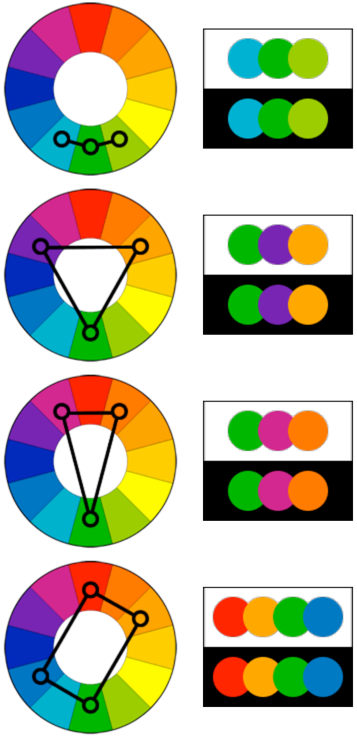


Complementary Colors

- Newton's arrangement of color on a circle
  - Based on the **painters** primary colors red, yellow, blue
- Arrangement of complementary colors opposite of each other
  - Complementary colors enhance the other's effect through optical contrast
- Secondary colors formed by mixing two primaries
  - Violet (R-B), green (B-Y), and orange (Y-R)
- Tertiary colors mixed from one primary and one secondary
  - Colors opposite one another on the wheel are complementary
  - Separated by two or three color positions are contrasting

Color Harmonies

- Analogous colors match well and create comfortable designs
  - Neighbors to one dominating color
  - Often found in nature
- Triadic colors are vibrant even if unsaturated
  - Evenly spaced around color wheel
  - Use one dominating and other two more sparsely
- Split-complementary
  - Two colors adjacent to the complement
  - Strong visual contrast but less obtrusive than direct complements
- Rectangle (tetradic)
  - Two complementary nearby pairs
  - Best with one dominant color



Cultural Aspects

Japan

”Ao” was used for different shades of blue including green

“Midori” was introduced later in the 1917, with the crayons.

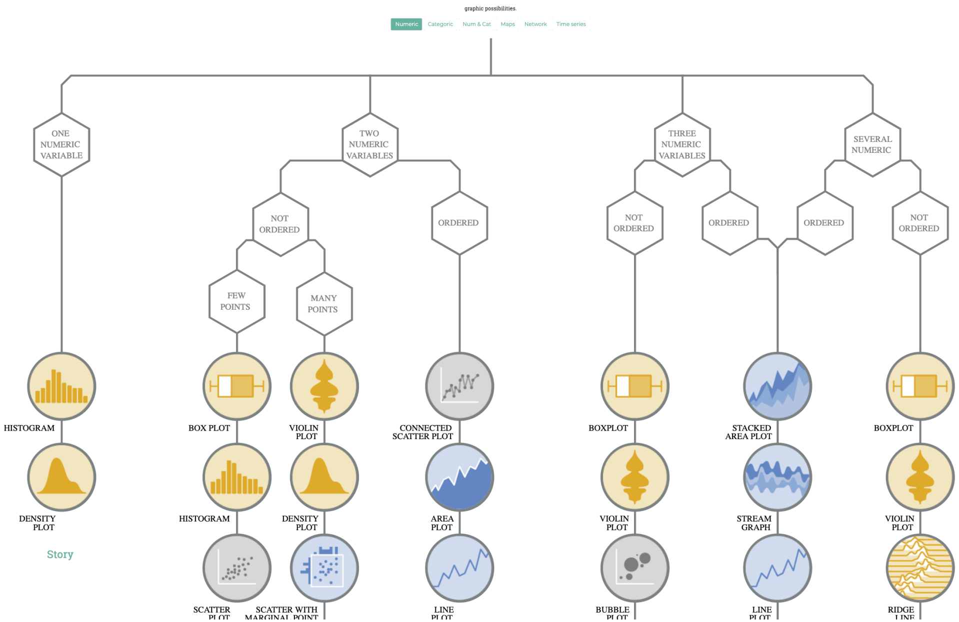
Is there any universal laws that can be apply to Color?

$$\begin{bmatrix}\text{white}\\\text{black}\end{bmatrix} < \begin{bmatrix}\text{red}\end{bmatrix} < \begin{bmatrix}\text{green}\\\text{yellow}\end{bmatrix} < \begin{bmatrix}\text{blue}\end{bmatrix} < \begin{bmatrix}\text{brown}\end{bmatrix} < \begin{bmatrix}\text{purple}\\\text{pink}\\\text{orange}\\\text{grey}\end{bmatrix}$$

A picture worth many words. The path to a more colorful language, according to Berlin and Kay (1969).

Key Aspects of Color Design

- What kind of data do I have?
- What task do I need to solve?
- Who is my user?



https://datavizproject.com/

All States

Choose A State ▾

Data Notes >

Daily New Cases per 100k people. Data shown from **1/22/20 to 11/18/22**.

Daily new cases (7-day moving average)

This page was last updated on Saturday, November 19, 2022 at 05:13 AM EST.

As states throughout the U.S. lift stay-at-home orders, reopen businesses, and relax social distancing measures, this graph shows whether cases of COVID-19 are increasing, decreasing, or remaining constant within each state.

Federal guidelines advise that states wait until they experience a downward trajectory of documented cases within a 14-day period before proceeding to a phased opening. Lifting social distancing measures prematurely, while cases continue to increase or remain at high levels, could result in a resurgence of new cases.

In this visualization, states that appear in shades of orange have experienced a growth in new cases over the past two weeks. States that appear in shades of green have seen declines in cases over the same period of time. The shade of the colors indicates the size of each state's growth or decline in new cases; the darker the shade, the bigger the change.

How to use this graphic:

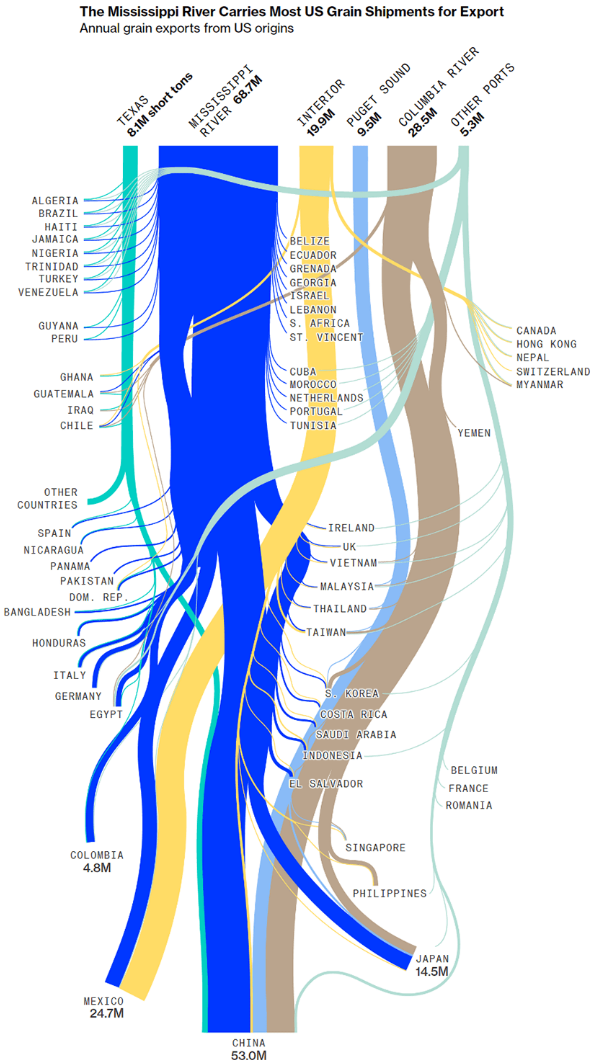
Click on a state to see more detail.

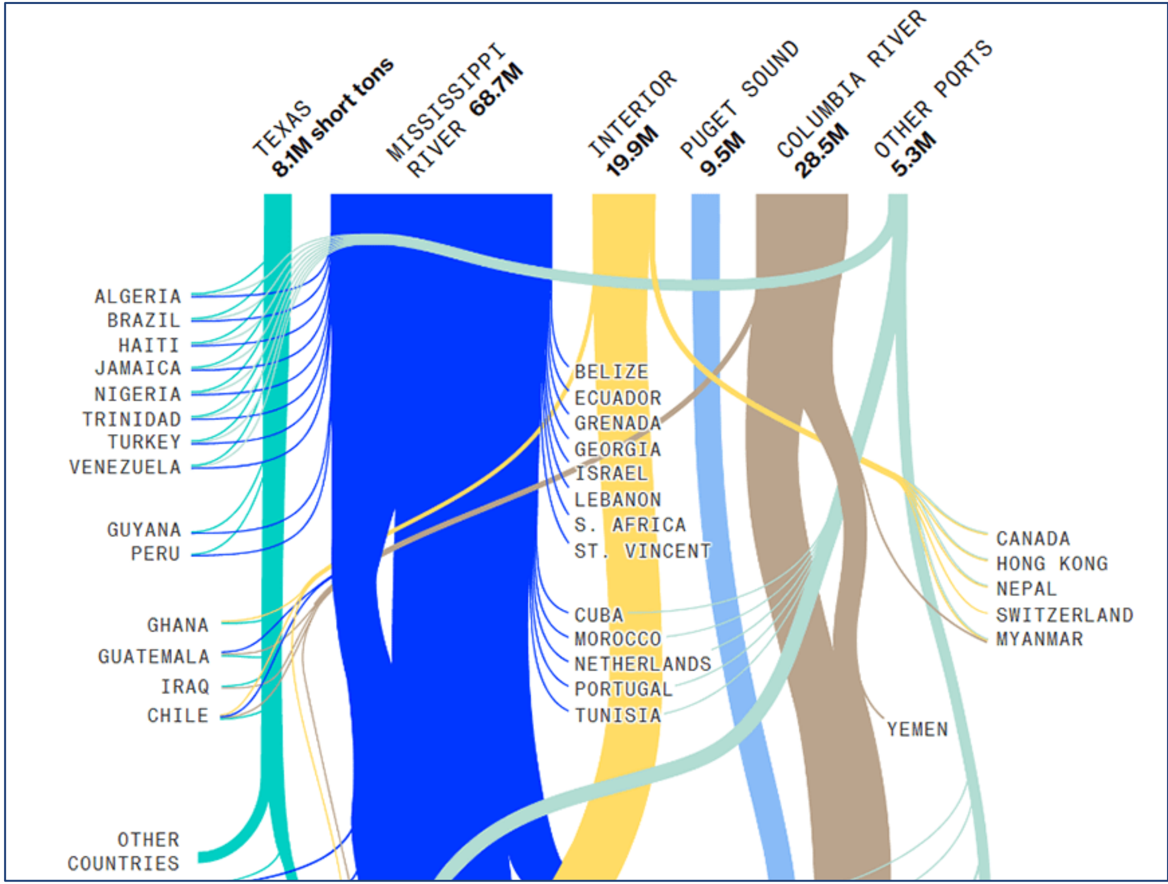
Line shows 7-day moving average of **new cases per day** in this state. Dot corresponds to most recent day.

The **greener** the background, the bigger the **downward trend** of new cases in this state.

The **redder** the background, the bigger the **upward trend** of new cases in this state.

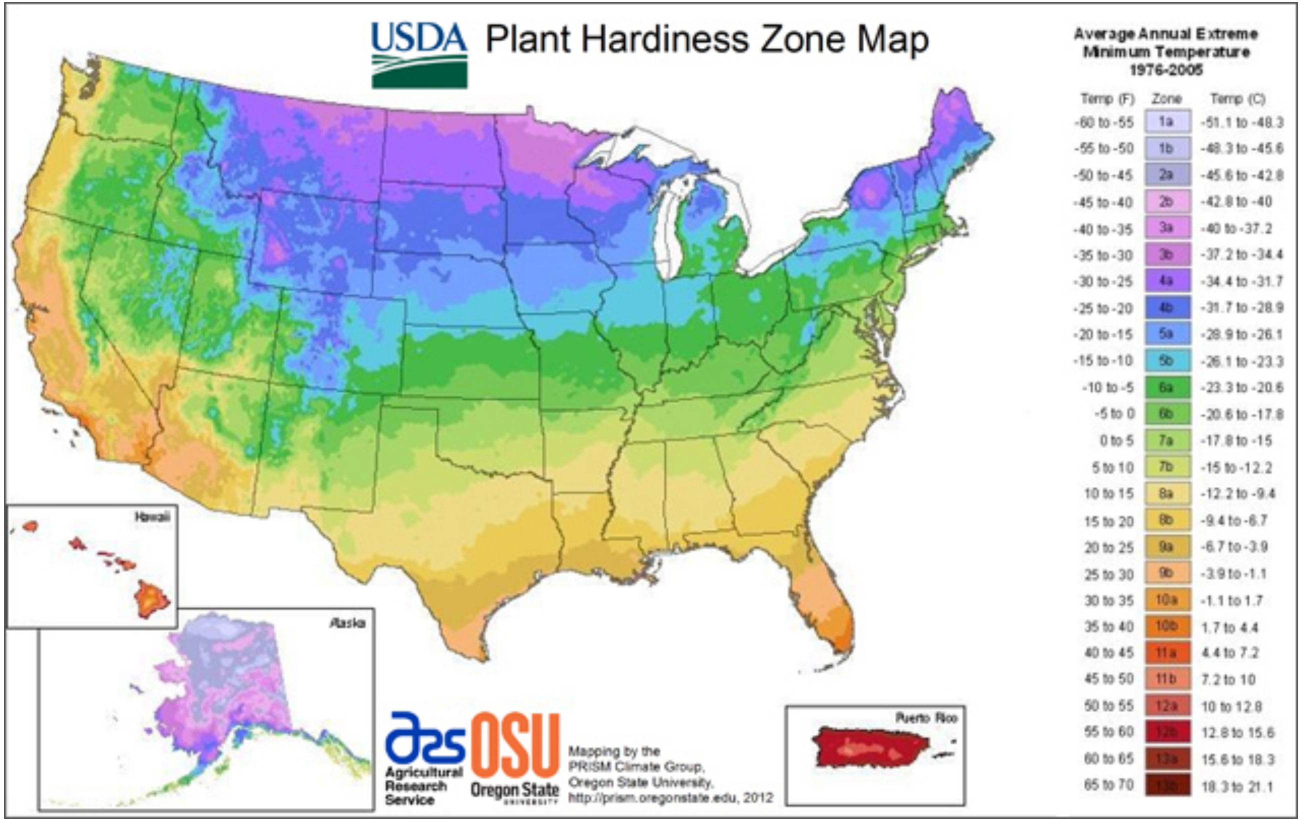
https://coronavirus.jhu.edu/data/new-cases-50-states





https://www.bloomberg.com/graphics/2022-mississippi-river-drought-global-impact/

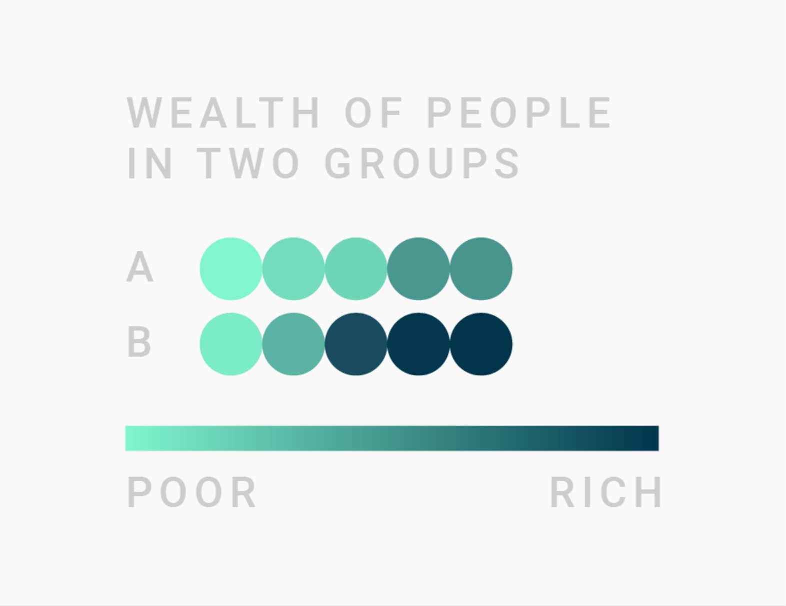
Good or bad color coding?



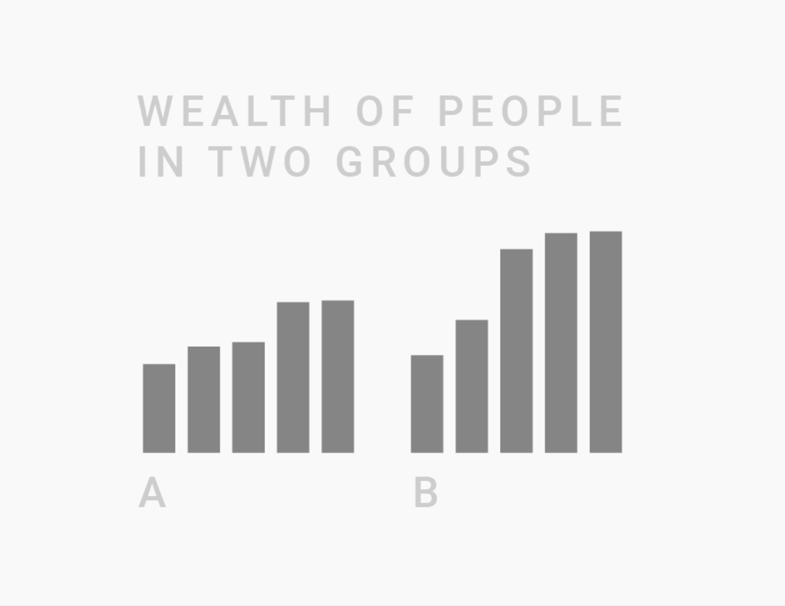
http://www.lawngrasses.com/images/maps/usda/usda-2012.jpg

How to make good color choices

consider other form of visual representation to encode most important values
make use of “position” first, before thinking about color



NOT IDEAL



BETTER

How to make good color choices

the more colors used in a chart, the harder it becomes to read
human ability to distinguish colors is limited to 12 (Ware, 2013)

NOT IDEAL

BETTER

How to make good color choices

reuse colors when creating multiple charts on the same data
(especially important for linked views)

NOT IDEAL

BETTER

How to make good color choices

explain what colors encode, i.e., **provide a legend**

DIFFERENT EXAMPLES OF COLOR KEYS

How to make good color choices

consider the use of “grey” for less important information, e.g., the context
use colors for selected items that stick out better

NOT IDEAL

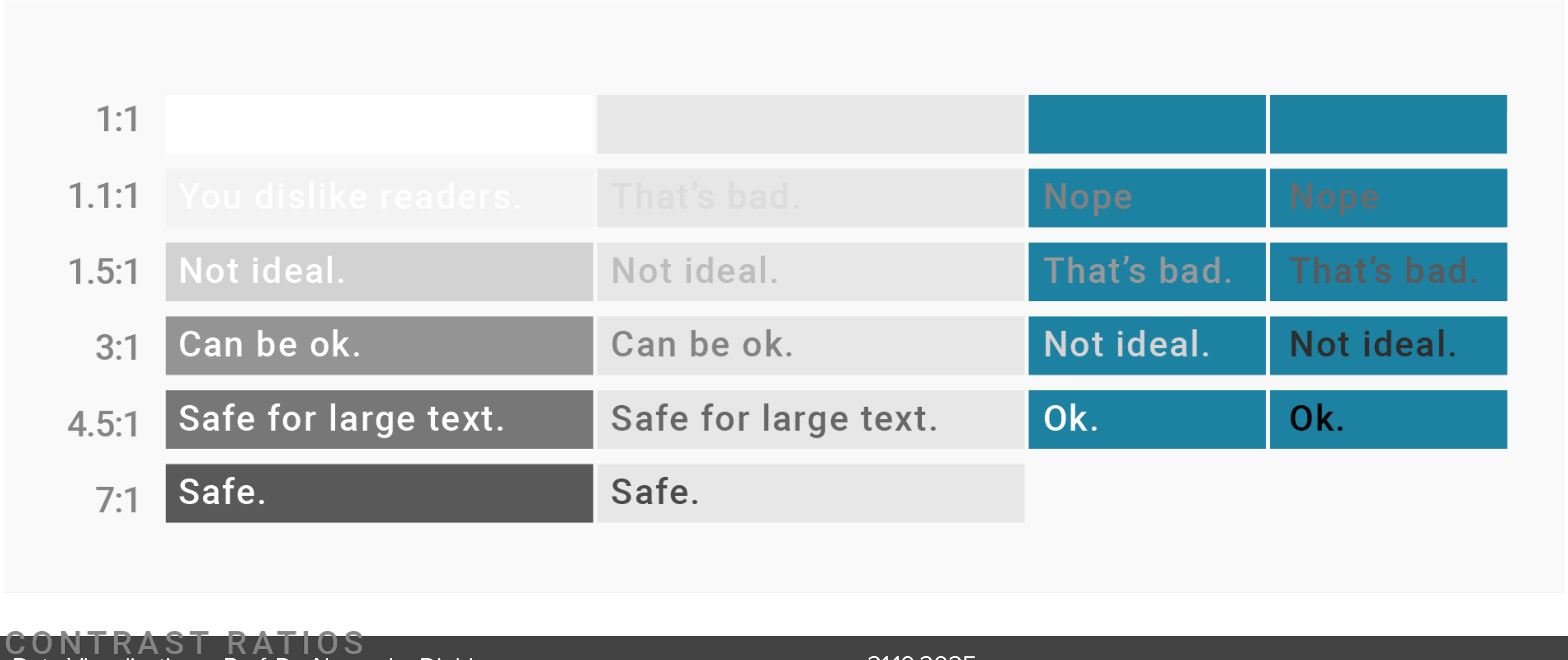
BETTER

How to make good color choices

choose an appropriate contrast to distinguish foreground and background

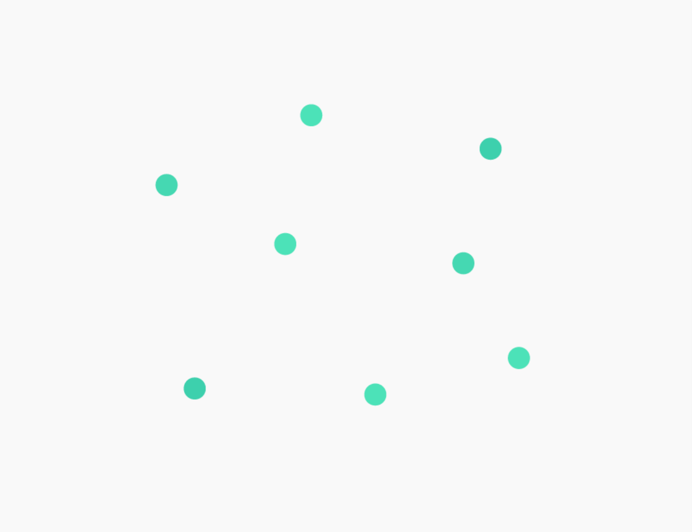
contrast ratio should be at least 2.5 for big text and at least 4 for small text

https://snook.ca/technical/colour_contrast/colour.html

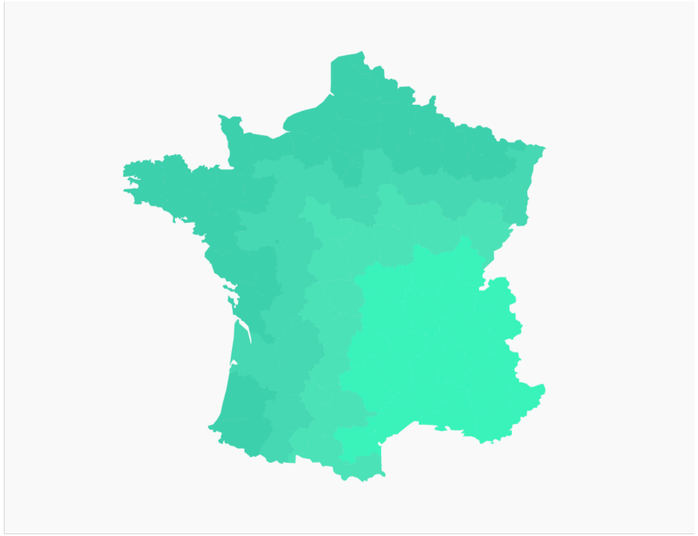


How to make good color choices

the smaller the areas on the chart and the bigger the distance between them,
the harder it is to compare them
small points or lines should have a high contrast in their hue or brightness



NOT IDEAL



BETTER

How to make good color choices

use intuitive colors, i.e., consider meaning of color in the culture of your target audience (e.g., party colors, natural colors), be careful when color-encoding gender data

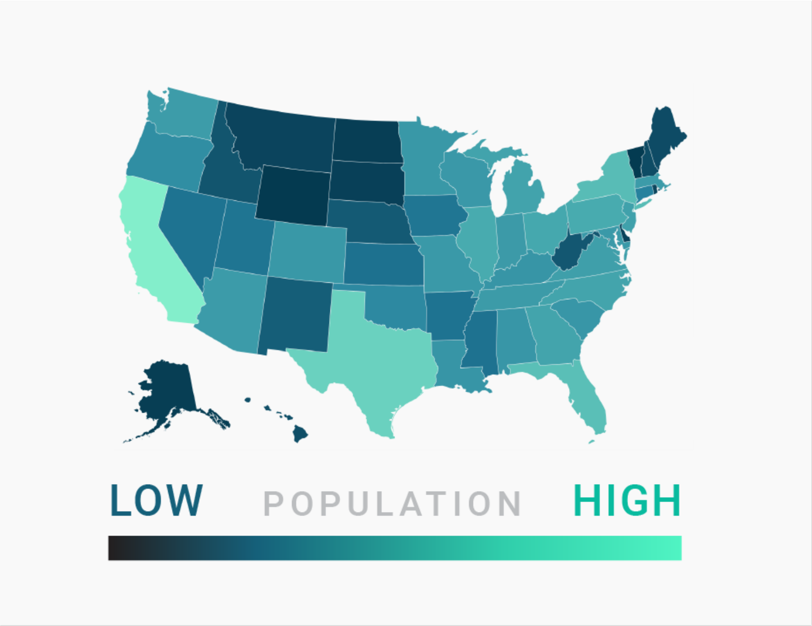
NOT IDEAL

BETTER

How to make good color choices

use color gradients appropriately

use light colors for low values and dark colors for high values



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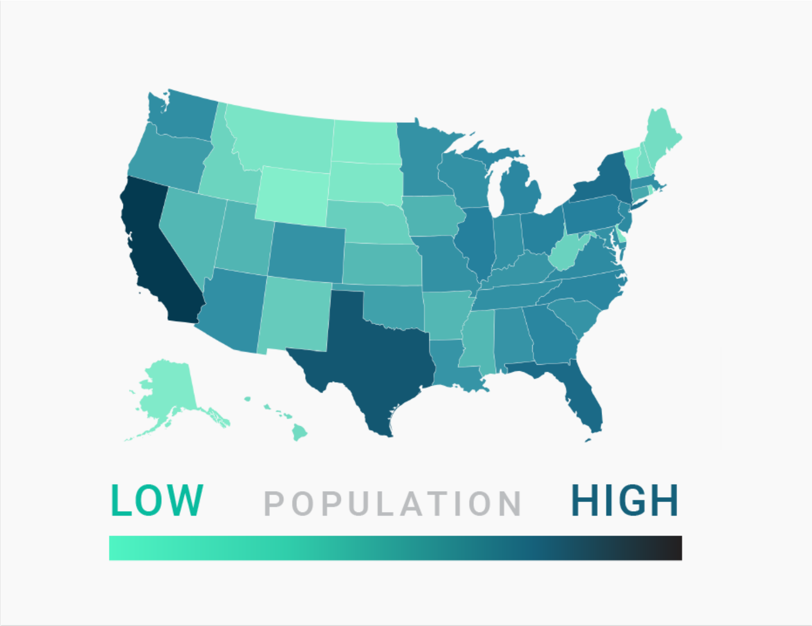


BETTER

How to make good color choices

choose the accurate color map

do not use a gradient color palette for categories

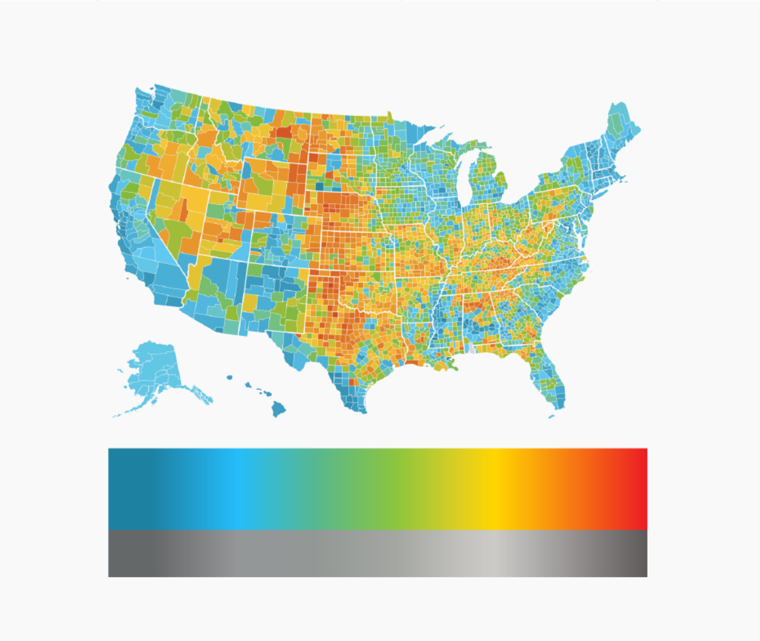
do not use categorical colors with diverse hue for quantities

NOT IDEAL

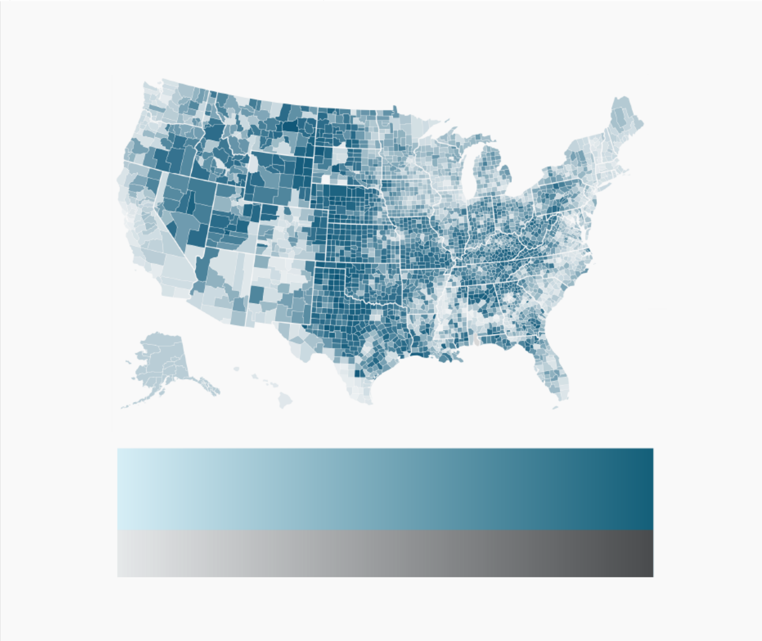
BETTER

How to make good color choices

gradient should not contain more than two hues with the same lightness
gradient should work in black and white
gradients with many variations in lightness (like rainbow scales) can confuse readers



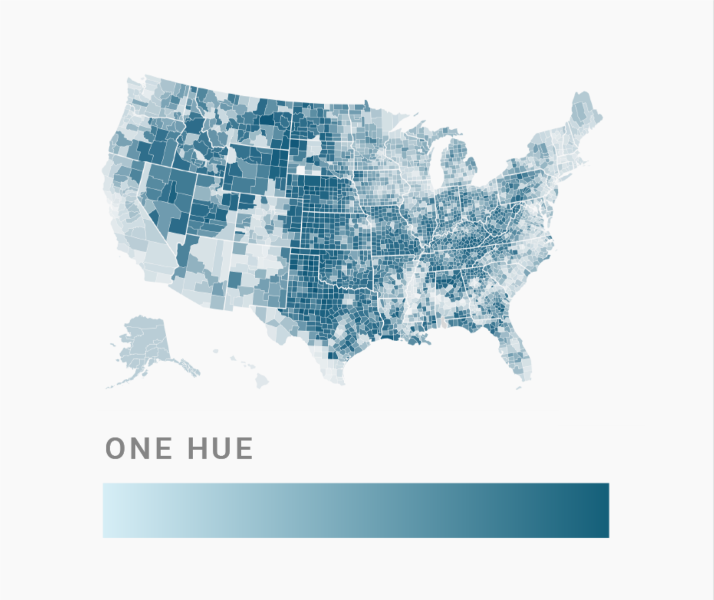
NOT IDEAL



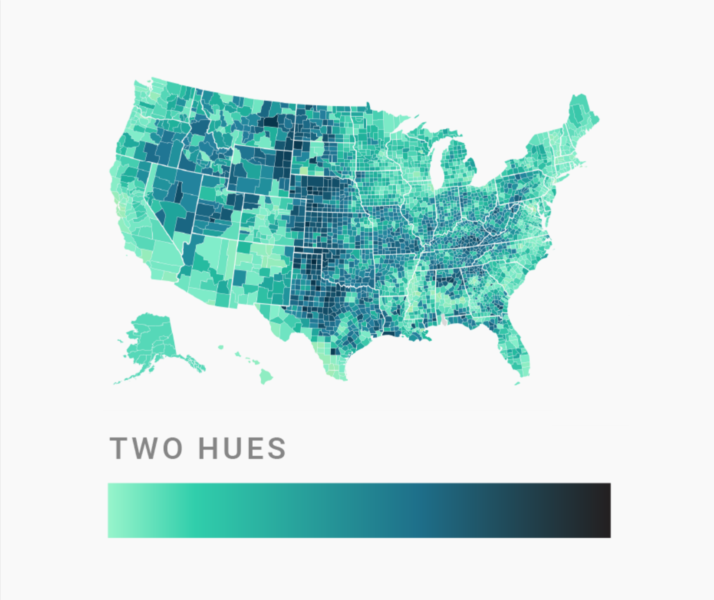
BETTER

How to make good color choices

carefully select multiple hues in gradients
can make maps better decipherable than just a single hue



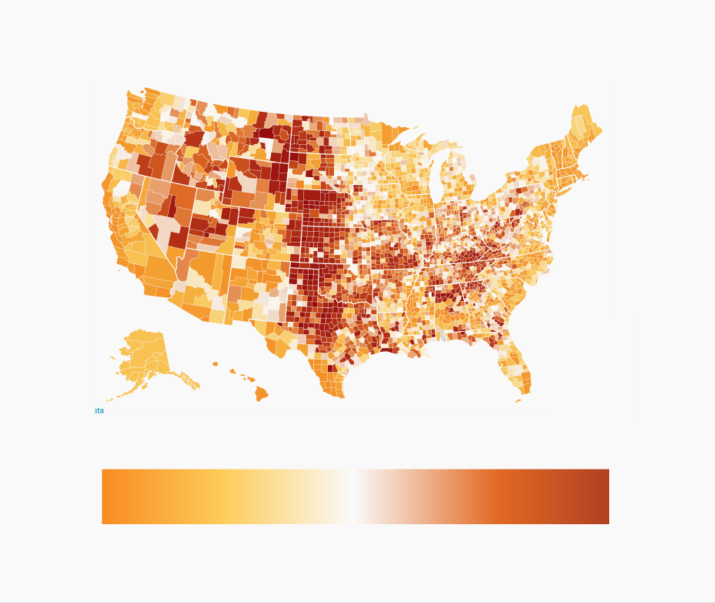
NOT SO BAD



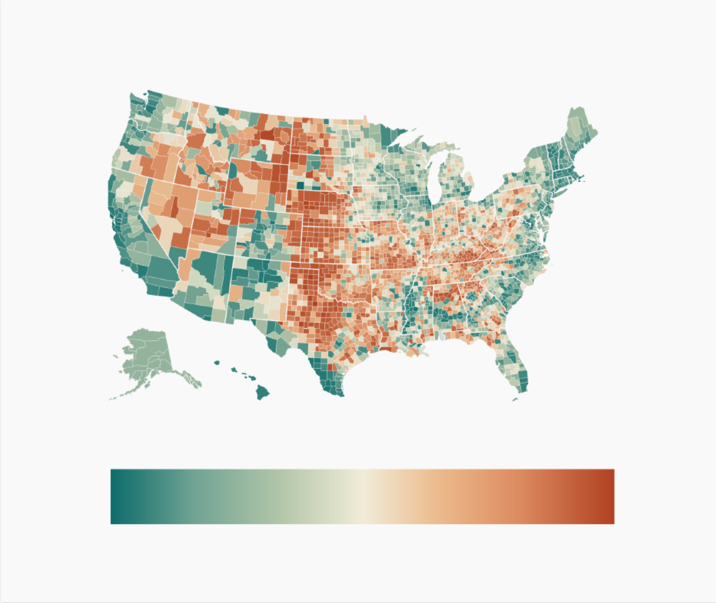
EVEN BETTER

How to make good color choices

consider using diverging color maps to show diverts from a baseline
again, distinguishable hues and grey (instead of white) in the middle



NOT IDEAL

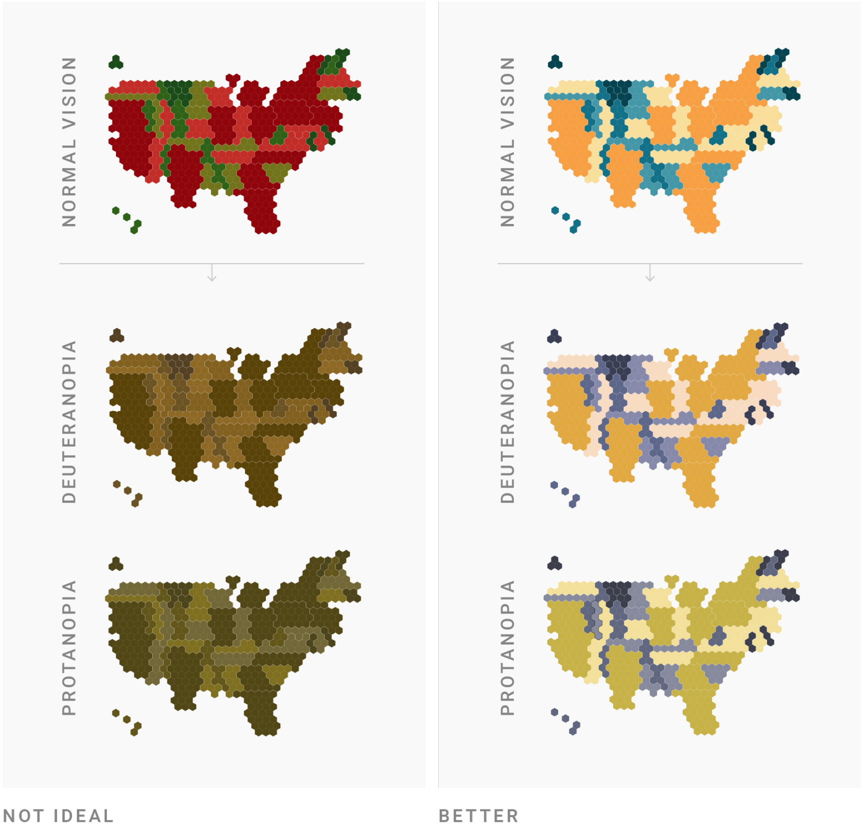


BETTER

How to make good color choices

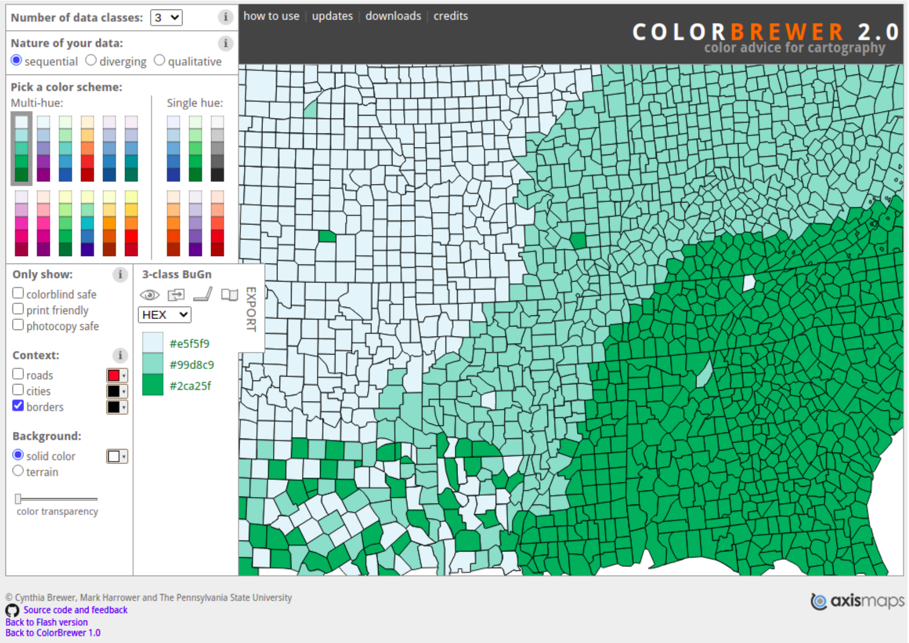
consider color-blind people
–> create color-blind friendly maps

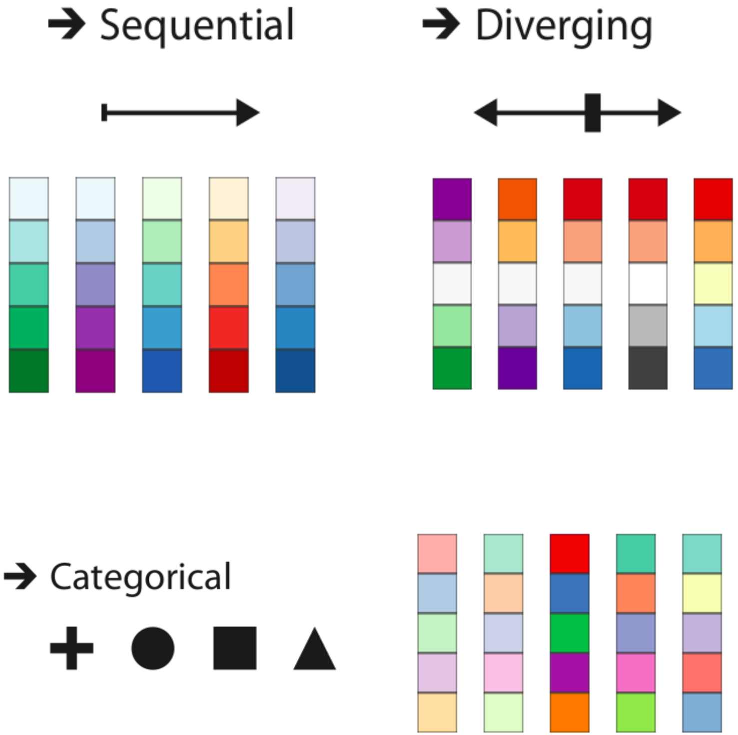
color blindness simulator: http://www.color-blindness.com/coblis-color-blindness-simulator/



Color mapping tools

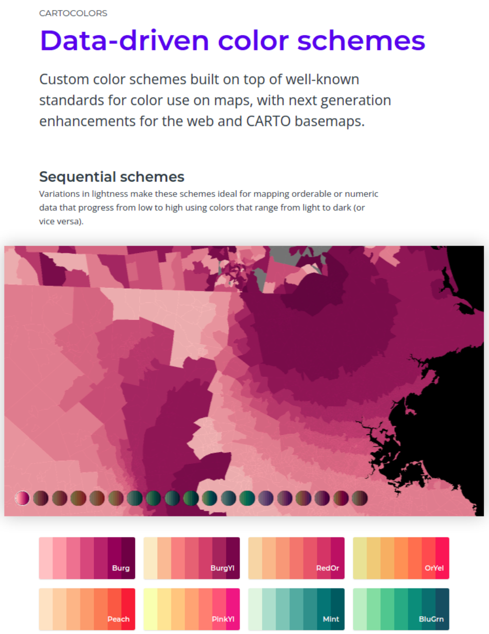
https://colorbrewer2.org/



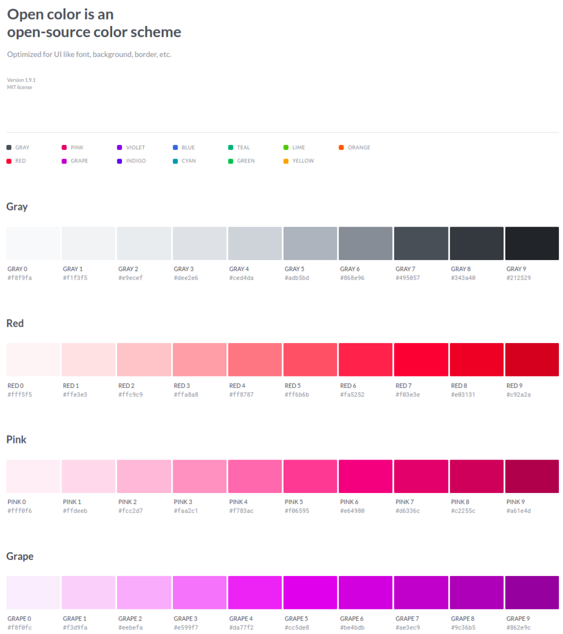


Color mapping tools

https://carto.com/carto-colors/



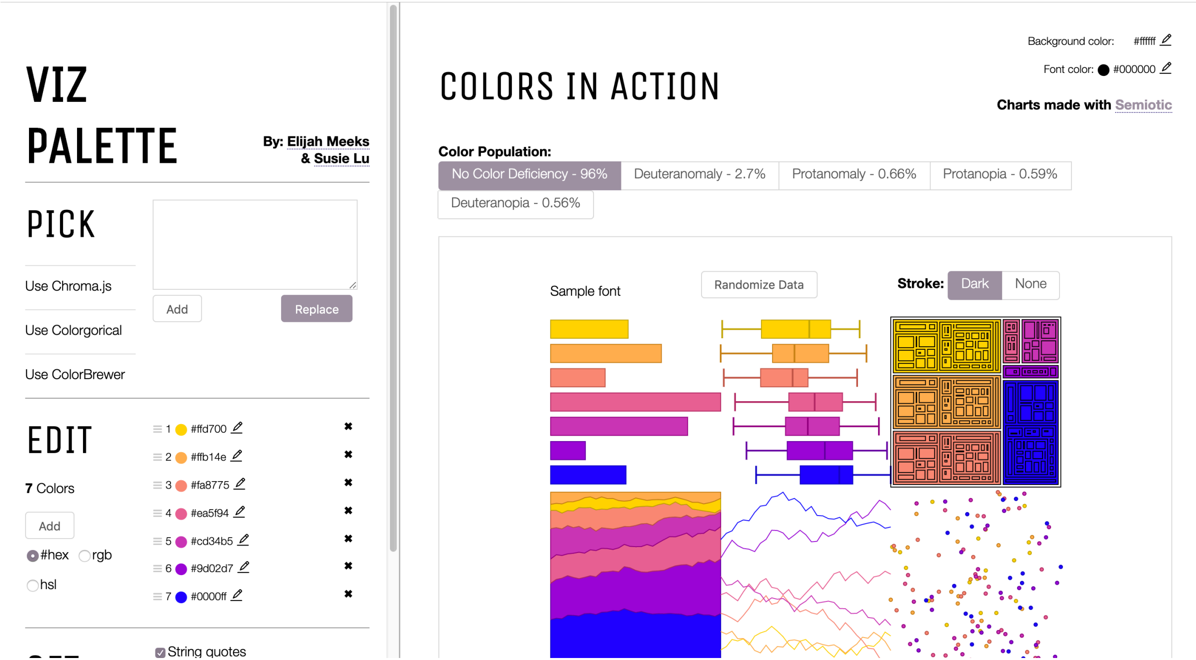
https://yeun.github.io/open-color/



Python/Plotly: https://plotly.com/python/colorscales/

JavaScript/D3: https://github.com/d3/d3-scale-chromatic

Color mapping tools



https://projects.susielu.com/viz-palette

Further Reading

Ware, C. (2019). Information Visualization: Perception for Design. Morgan Kaufmann.

Harrower, M., & Brewer, C. A. (2003). ColorBrewer. org: an online tool for selecting colour schemes for maps. The Cartographic Journal, 40(1), 27-37.

Zhou, L., & Hansen, C. D. (2015). A survey of colormaps in visualization. IEEE transactions on visualization and computer graphics, 22(8), 2051-2069.

Szafir, D. A. (2017). Modeling color difference for visualization design. IEEE transactions on visualization and computer graphics, 24(1), 392-401.

What to consider when choosing colors for data visualization?
https://academy.datawrapper.de/article/140-what-to-consider-when-choosing-colors-for-data-visualization